Lakshmi Ravuri

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Professional Experience

Software Developer - *New jersey Department of Transportation* Jun 2024 – Present

- Designed and implemented **Python-based ETL pipelines** for ingesting, cleaning, and transforming multi-gigabyte field sensor datasets, improving processing speed by **40%** and enabling near-real-time analytics.
- Developed **RESTful APIs** in **Django** to deliver structed and aggregated pavement and skid management data to multiple systems, reducing manual data preparation time by **over 60%.**
- Created **interactive dashboards** with Plotly.js, D3.js, and HTML/CSS/Javascript, allowing engineers to visualize and filter road condition metrics with zero page reloads.
- Automated database operations with Python scripts integrating **Oracle** and **PostgreSQL**, improving data consistency and reducing QA-related rework by **30%.**
- Deployed containerized microservices in **Docker** and **Azure Kubernetes Service** (AKS) for scalability and reliability.

Research Engineer - Machine Learning – *Rowan University* Sep 2022 - May 2024

- CNN and LSTM models applied structural sensor datasets to detect civil infrastructure failures with **90%+ accuracy**.
- Built ML models (data preprocessing, training, inference), enabling **real-time anomaly detection** on smart bridges.
- Conducted research on **ML techniques** for structural monitoring, specializing in analyzing **time-series,** and **point cloud data.**
- Deployed ML models on edge systems and visualized outputs using interactive Plotly.js dashboards for engineers and researchers.
- Developed **hybrid models** using sensor data to determine damage in reinforcing bars inside concrete bridge columns.
- Implemented **NLP pipelines** for text classification tasks, reducing manual grading workload by **50%**.
- Mentored **junior researchers** and collaborated with DOT officials, bridging the gap between **AI research** and **infrastructure safety.**

Software Engineer - *Anthology* Apr 2021- Jul 2022

- Engineered scalable **Java microservices** with **Spring Boot**, boosting **API response times by 35%** and supporting **10K+ daily users**.
- **Architected and optimized CI/CD pipelines** with **Jenkins** and **Maven**, accelerating **deployments by 40%** and minimizing **downtime**.
- Developed robust **unit and integration test suites JUnit** and **Mockito**, raising **code coverage to 90%** and reducing **production defects by 25%**.
- Collaborated closely with **frontend teams** to implement **REST APIs** that streamlined **data flow**, enhancing **user experience** and reducing **frontend load times by 20%**.
- Led **Agile sprint planning** and **code reviews**, driving consistent delivery of **high-quality features** aligned with **business priorities**.

Skills & Abilities

- **Languages:** Java, Python, SQL, TypeScript, JavaScript, HTML/CSS, VBA, C, R, MATLAB
- **Frameworks & Libraries:** Spring Boot, Fast API, React.js, Node.js, Django
- **Databases:** Oracle, PostgreSQL, MYSQL, MongoDB, DynamoDB, SQL Query Optimization
- **CI/CD & Tools:** Git, GitHub, Jenkins, Docker, VS Code, IntelliJ, Eclipse, Jira, Confluence, Linux, YAML
- **Visualization:** Plotly.js, D3.js, Tableau, Power Bl, Dash
- **ML & Data Science:** CNN, LSTM, PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, Fast API, SHAP, Grad-CAM, NLTK, LLMs, NLP
- **Other:** VBA Shell Scripting, LaTeX, SOAPUI, SQL Query Optimization, Agile & Scrum, RESTful APIs, Selenium, Playwright, Appium, TestNG, JUnit, Postman, Newman, Jenkins, GitHub Actions, BDD, Cucumber; POM, Xray, Applitools Eyes, Gherkin

Publications

- **Journal: Structures** Predicting low-cycle fatigue-induced fracture in reinforcing bars: A CNN-based approach view
- **IABSE, Manchester Conference:** Predicting Fractures in Reinforcing Steel Bars: A Low Cycle Fatigue CNN Approach view
- **WCEE, Milan Conference** Machine Learning based SHM of Rocking Bridge System under Seismic Excitations view
- **Publisher: Elsevier** Harnessing Data from Benchmark Testing for the Development of Spalling Techniques Using DL view
- **Rowan Digital Works:** Integration of ML in SHM for Damage Identification and Response Prediction in Bridges view

Education

Master of Science in Data Science, (Rowan University, Glassboro, NJ, USA)
GPA: 4.0/4.0 - *Thesis Track – Golden Key International Honors Society*
Bachelor of Science in Aeronautical Engineering, (Sathyabama, Chennai, India)
GPA: 9.69/10.00 - *Gold Medalist*